Design and Implementation of a Non-coupled Integrated Transformer for a Two-phase Interleaved Flyback PFC Rectifier for USB PD 3.1 Applications

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***Abstract*—Universal serial bus power delivery (USB PD 3.1) is a widely-applied standard for electronic device charging. In 2021, USB PD 3.1 was announced with a variable output of 5 V to 48 V and a maximum output power of 240 W. A multi-phase flyback rectifier is capable of power factor correction, galvanic isolation, and wide output voltage range, and is proposed as an alternative topology for USB PD. However, the number of magnetic components increases with the number of phases, which can lead to lower power density. This paper proposes the design of a non-coupled integrated transformer for a two-phase interleaved flyback rectifier to reduce the size and number of magnetic components. The independent operation of the integrated transformer in two phases was verified using a 240-W two-phase interleaved flyback PFC rectifier at 100 kHz switching frequency operating at discontinuous conduction mode (DCM), both in simulation and experiment. After conducting a volume analysis through the area product method, the proposed transformer is up to 19% smaller compared to its conventional counterpart.**

***Index Terms*—Integrated magnetics, Transformer design, Two-phase interleaved flyback rectifier, USB PD 3.1, PFC, Core volume analysis.**

I. Introduction

UNIVERSAL serial bus power delivery (USB PD) is a technology used for convenient, portable, fast charging in electronic devices. The latest standard, USB PD 3.1, was announced in 2021, with the maximum power supplied being up to 240 W, allowing for higher power applications [1]. Adapters for USB PD 3.1 must be capable of power factor correction (PFC) for supporting 75 W and above, be equipped with galvanic isolation for user safety, and be capable of providing an output voltage of 5 V to 48 V [2].

A flyback rectifier has been widely used for commercial adapters due to its simple configuration, easy PFC control, and wide output voltage range. Improved configurations of flyback rectifiers such as active clamp flyback converters [6], [7] and bridgeless flyback rectifiers have also been proposed for higher converter efficiency [8]–[11]. However, they are generally more practical for lower power applications due to the inherent high current stress on the switches by these topologies.

Several power supply configurations for USB PD 3.1 have been proposed to meet the requirements. One example is a combination of a PFC rectifier, an LLC converter, and a buck converter as shown in Fig. 1(a) [3], [4]. While this configuration achieves high power density, the additional buck stage required to compensate for the limited gain range of the LLC increases the total component count and system cost. An asymmetrical half bridge (AHB) flyback converter in the second stage of a boost PFC circuit was introduced as shown in 1(b) [5]. The AHB allows for zero-voltage switching (ZVS) by the application of resonant capacitors in series placed on the primary side of the transformer. As a single-phase topology, the AHB processes the entire load power through a single transformer and switch pair. While this minimizes the component count, it inherently concentrates thermal stress in specific circuit elements.

Interleaving rectifiers have been considered to increase the maximum output power. Two-phase interleaving allows half of the total power to be supplied in each phase, therefore reducing the possible stress on the circuit elements [12]. The topology is shown in Fig. 1(c), where the two phases are parallel in the input and output side. The source nodes of switches S_1 and S_2 are common, and only one power supply is required for gate drive circuits. A common cathode package can be used for diodes D_1 and D_2. Fewer components can configure the circuit though two transformers are required. However, adopting two transformers compromises the power density of the flyback rectifier.

Integrated magnetics have been widely utilized to minimize component size and improve power density. The transformer in [15] features one set of primary and secondary windings placed on each of the outer legs of a three-legged core. This arrangement does not allow for flux cancellation in the center leg and is designed such that both transformers operate simultaneously. While this structure may help reduce input capacitor loss by distributing the ripple current, it requires precise tuning of the input current ripple and coupling, adding complexity to the design. The transformer in [16] adopts a similar physical layout as in [15], but with the windings arranged to enable flux cancellation in the center leg. Coupling between the windings allows for a reduction in input current ripple; however, the design becomes more complex, since the

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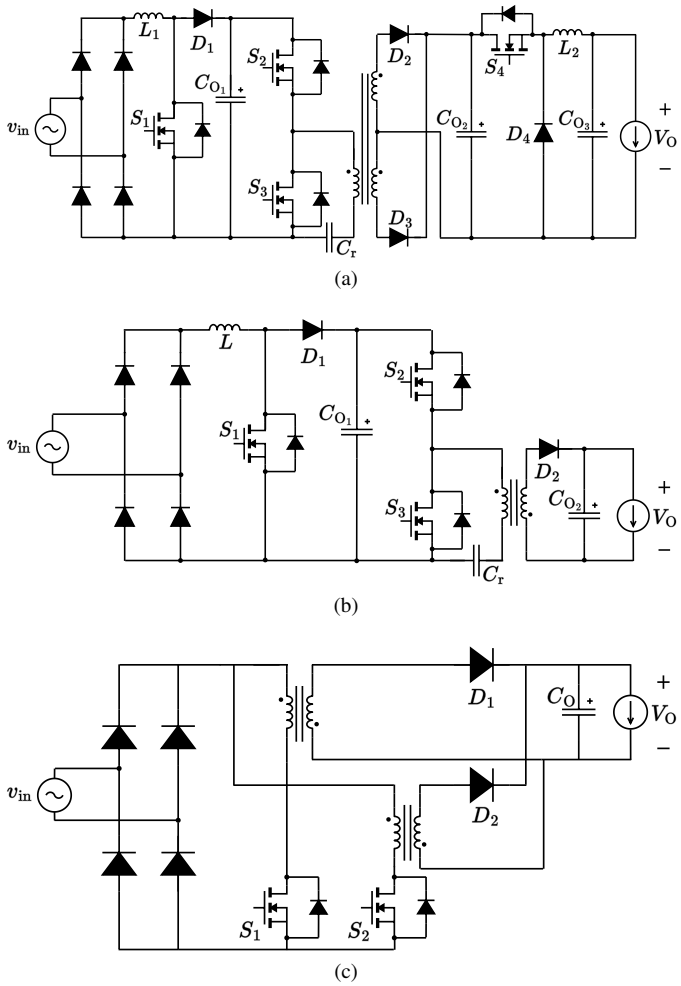


Fig. 1. Power supply configurations proposed for PD 3.1 applications: (a) PFC boost, LLC, and buck converter combination in [3], [4], (b) PFC boost with AHB flyback [1], (c) two-phase interleaved flyback PFC rectifier.

magnetic coupling affects zero-voltage switching (ZVS) and active clamping. The transformer in [17] also uses a similar configuration, but with inversely coupled secondary windings. In this case, careful design of the secondary-side coupling is required to meet specific output current ripple requirements, and the inverse coupling may introduce unwanted circulating currents or EMI if not properly managed. The transformer in [18] adds an inductor winding on the center leg, in addition to the transformers on the outer legs, to further control output current ripple. However, this increases the core's magnetic complexity and can make magnetic design and analysis more challenging. A detailed comparison of these integrated magnetics structures is provided in Table I.

This paper proposes a non-coupled integrated transformer for a two-phase interleaved flyback rectifier as shown in Fig. 2(a). To ensure independent phase operation without magnetic interference, the structure is designed to achieve zero magnetic coupling between phases. The proposed winding structure is shown in Fig. 2(b) where N_{12} is the number of turns between terminals 1 and 2. Half of N_{12} and N_{34} are wound at each outer leg with opposite polarity, so as to allow for flux cancellation at the center leg. On the other hand, N_{56} and N_{78} are also wound with opposite polarity on the center leg, and allows for the transformer N_{12} and N_{34} to not operate during its operation.

The zero-coupling coefficient between the two separate transformers simplifies implementation of rectifiers and transformer design. The operation of the rectifiers in each interleaved phase still follows that of a conventional flyback converter, preserving simplicity in control and timing. The design is also highly dependent on the duty cycle, an easily adjustable parameter in circuit design. The converter in this study aims to operate in DCM in order to allow for easier implementation of PFC and to conform to the standards of USB PD 3.1. It also realizes independent phase operation, allowing the circuit to function in one-phase mode if desired, thereby enhancing flexibility in circuit operation.

The rest of this paper is organized as follows: Section II presents an operational analysis of a two-phase interleaved flyback rectifier with the integrated transformer, along with its design method. Section III describes the size reduction effect of the transformer. Sections IV and V discuss the results of the simulation and experimental verification respectively, and Section VI presents the conclusions of this paper.

II. Design Method of the Proposed Transformer

A. Operation Analysis of the Proposed Flyback Rectifier and Transformer

The proposed integrated transformer, shown in Fig. 2(b) was designed for a two-phase interleaved flyback PFC rectifier, shown in Fig. 2(a) operating in discontinuous conduction mode (DCM). The switches (S_1, S_2 in Fig. 2(a)) operate with a fixed duty cycle D and a 180-degree phase shift.

Half of N_{12} and N_{34} are wound at each outer leg with opposite polarity. Moreover, N_{56} and N_{78} are also wound with opposite polarity at the center leg. The transformer utilized an EE Core with a center leg that has a cross sectional area twice

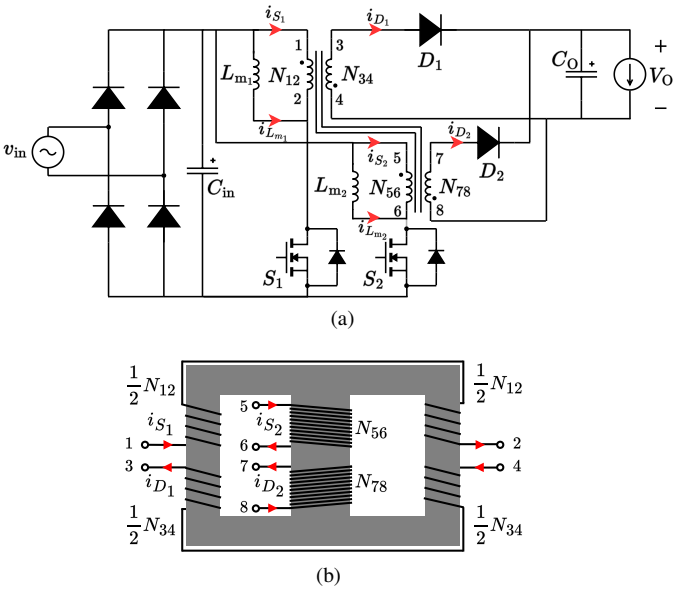
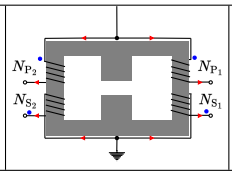
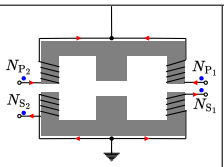
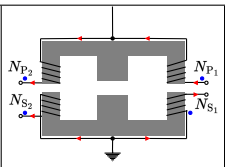
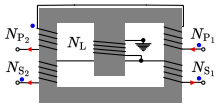
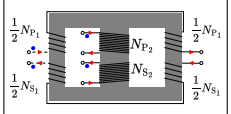


Fig. 2. The two-phase interleaved flyback rectifier (a), with the transformer and its winding structure in (b).

TABLE I
COMPARISON OF INTEGRATED MAGNETICS TECHNIQUES WITH THE PROPOSED TRANSFORMER

					
Transformer	[15]	[16]	[17]	[18]	**Proposed Transformer**
Circuit Topology	Two-phase Interleaved Flyback	Two-phase Interleaved Forward	Two-phase Interleaved Flyback	Full-wave converter	**Two-phase Interleaved Flyback**
Operation	CCM, step-up	CCM, step-down	CCM, step-down	CCM, step-down	**DCM, step-down**
Transformer Configuration	Series-input, Parallel-output	Series-input, Parallel-output	Series-input, Parallel-output	Parallel-input and output	**Parallel-input and output**
Coupling between Transformers	Yes	Yes	Yes (inverse coupling)	Yes	**No**
Flux Cancellation in Middle Leg	No	Yes	No	-	**Yes**
Air Gap	Center leg	Outer and Center Legs	Outer and Center Legs	-	**Outer and Center Legs**
Feature of Circuit with Transformer	Less input capacitor losses due to reduction of input current ripple.	Input current ripple reduction	Output current ripple reduction	Output current ripple controllable	**2 phase independent operation, 1 phase drive possible.**
Design Method	Uses primary side current ripple and peak flux (affected by coupling coefficient)	Transformer designed with accordance to implementation of ZVS and Active Clamping.	Uses primary side current ripple, saturation magnetic flux density, and coupling coefficient	Uses primary side current ripple, and saturation magnetic flux density	**Uses duty cycle and saturation magnetic flux density**

its outer leg, as per commercial standard. The magnetizing inductances for both transformers $N_{12}-N_{34}$ and $N_{56}-N_{78}$ were set to be equal, and are denoted as L_{m_1} and L_{m_2} respectively. The saturation magnetic flux in each separate transformer during each phase is expected to be consistent.

The operational principle of the rectifier with the proposed transformer is illustrated in Fig. 3. Additionally, Fig. 4 presents the key waveforms of the six distinct operation modes. The rectifier is designed with duty cycle $D \leq 0.5\frac{nV_{\mathrm{o}}}{V_{\mathrm{in}_{\max}}}$ with n as the turns ratio, V_{o} as the output voltage, and $V_{\mathrm{in}_{\max}}$ as the peak value of the input voltage. This condition ensures the magnetizing current of one phase returns to zero before the other phase peaks. This duration is defined as the discharge time t_{dis} and is given in terms of circuit parameters by (1):

$$t_{\mathrm{dis}} = \frac{V_{\mathrm{in}_{\max}} D T_{\mathrm{s}}}{n V_{\mathrm{o}}} \tag{1}$$

with T_{s} as the switching period. For the condition given by D in this analysis, t_{dis} corresponds to the interval between t_3 and t_5 in Fig. 4.

To describe the magnetic state of the core, the superposition principle is used, as shown in Fig. 5(a). The magnetic flux density component B' is generated by the positive cycle transformer defined as $N_{12}-N_{34}$. The magnetic flux density component B'' represents the component generated by the negative cycle transformer defined as $N_{56}-N_{78}$. The total flux in any leg is the algebraic sum of these components. This superposition determines the peak flux density required for the core design, which is analyzed in detail in Section II-B.

Modes 1, 2, and 3 in Figs. 3(a)-(c) cover the operation during the positive AC input into the rectifier, while Modes 4, 5, and 6 in Figs. 3(d)-(f) describe the operation during the negative AC input. For the following modes, B'_{o_1}, B'_{o_2} and B'_{c} represent the magnetic flux densities in the respective left, right, and center legs of the core caused by the transformer active in positive AC input $N_{12}-N_{34}$, respectively, which are shown visually in Fig. 5(b). On the other hand, B''_{o_1}, B''_{o_2} and B''_{c} represent the magnetic flux densities in the respective left, right, and center legs of the core caused by the transformer active in negative AC input $N_{56}-N_{78}$, respectively, which are detailed visually in Fig. 5(c).

The detailed analysis of each operating mode is as follows:

1) Mode 1 ($t_1 < t < t_2$, Fig. 3(a)): At time t_1, switch S_1 turns on and S_2 remains off. The transformer $N_{12}-N_{34}$ begins storing energy in its magnetizing inductance. Its magnetizing current $i_{L_{\mathrm{m}_1}}$ and the corresponding flux densities in the outer legs B'_{o_1} and B'_{o_2} linearly increase from zero. Simultaneously, transformer $N_{56}-N_{78}$ completes its discharge phase, with its magnetizing current $i_{L_{\mathrm{m}_2}}$ and secondary diode current i_{D_2} decreasing to zero at t_2.
2) Mode 2 ($t_2 < t < t_3$, Fig. 3(b)): Mode 2 starts right after transformer $N_{56}-N_{78}$ completes its discharging phase, which means its associated currents i_{D_2}, $i_{L_{\mathrm{m}_2}}$, and flux densities B''_{o_1}, B''_{o_2}, and B''_{c} are zero. Now, with S_1 still on, transformer $N_{12}-N_{34}$ continues storing energy until $i_{L_{\mathrm{m}_1}}$ reaches its peak value at t_3. The peak primary current in the positive cycle $i_{\mathrm{S}_{1_{\mathrm{pk}}}}$ is given by (2):

$$i_{\mathrm{S}_{1_{\mathrm{pk}}}} = \frac{V_{\mathrm{in}} \cdot D T_{\mathrm{s}}}{L_{\mathrm{m}_1}} \tag{2}$$

where V_{in} is the input voltage, D is the duty cycle,

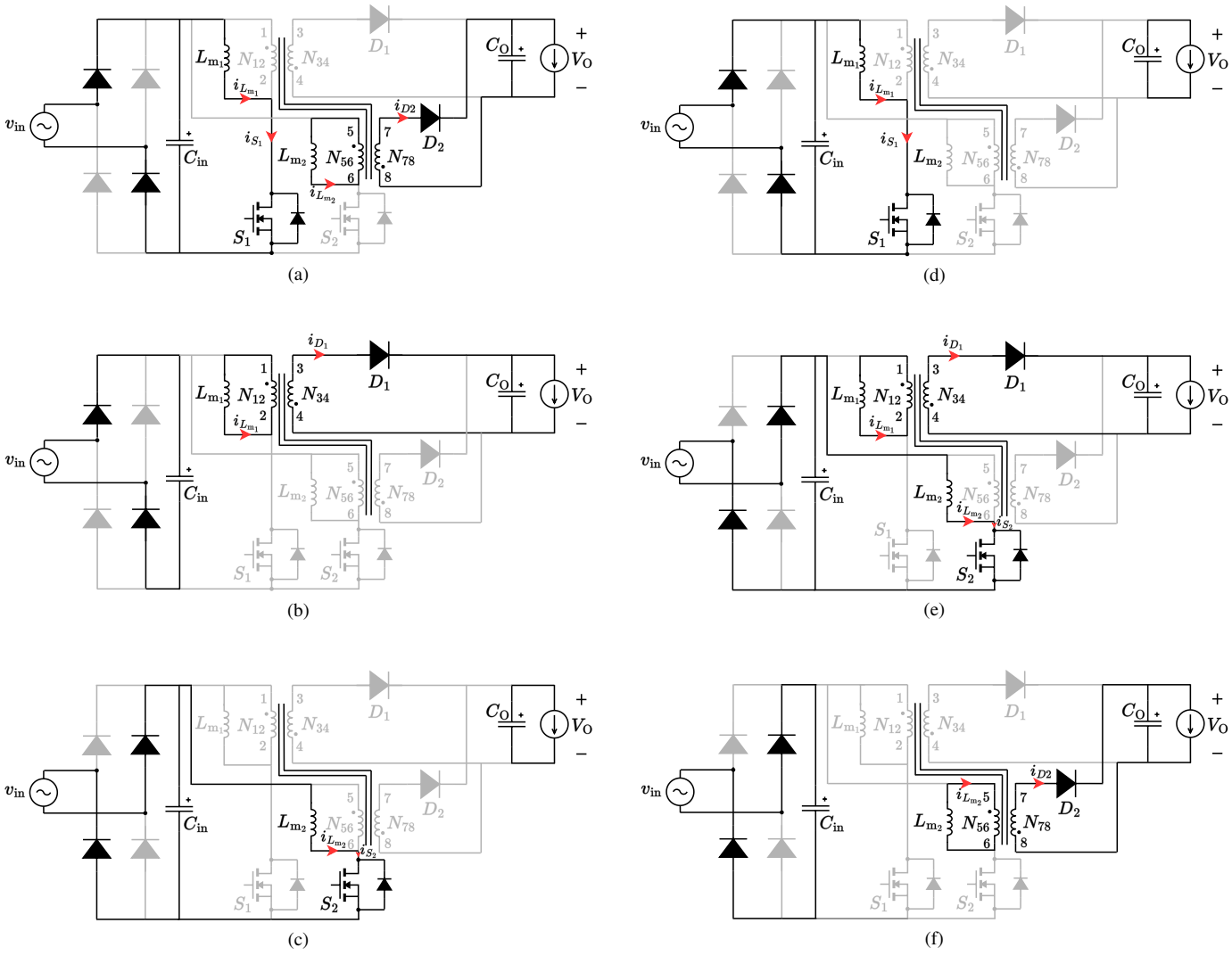


Fig. 3. Operational principle of the two-phased interleaved flyback rectifier with the proposed integrated transformer at $D \leq 0.5\frac{nV_o}{V_{in_{max}}}$. There are six distinct operational modes shown: (a) Mode 1 ($t_1 < t < t_2$), (b) Mode 2 ($t_2 < t < t_3$), (c) Mode 3 ($t_3 < t < t_4$), (d) Mode 4 ($t_4 < t < t_5$), (e) Mode 5 ($t_5 < t < t_6$), and (f) Mode 6 ($t_6 < t < t_7$).

T_s is the switching period, and L_{m_1} is the magnetizing inductance of transformer $N_{12} - N_{34}$. Consequently, the flux densities B'_{o_1} and B'_{o_2} also reach their peak values $B'_{o_{pk}}$, which are given by (3):

$$B'_{o_{pk}} = \frac{L_{m_1} i_{S_{1_{pk}}}}{N_{12} A_o} \tag{3}$$

with N_{12} as the number of turns in the primary side of the active transformer and A_o is the cross-sectional area of the outer leg, where the transformer $N_{12} - N_{34}$ is wound.

3) Mode 3 ($t_3 < t < t_4$, Fig. 3(c)): Mode 3 starts right after transformer $N_{12} - N_{34}$ finishes charging. At t_3, Switch S_1 turns off, initiating the discharge phase of transformer $N_{12} - N_{34}$. The magnetizing current $i_{L_{m_1}}$, diode current i_{D_1}, and flux densities B'_{o_1} and B'_{o_2} begin to decrease linearly. Transformer $N_{56} - N_{78}$ remains inactive in this period.

In Modes 1-3, the magnetomotive force and magnetic resistance of both outer legs are equal. The magnetic flux in the center leg is canceled. This causes no electromotive force generated in transformer $N_{56} - N_{78}$. Therefore, the transformer formed by the outer legs operates independently from the transformer in the center leg. The integrated transformer is equivalent to that of Fig. 5(b) and its equivalent magnetic circuit is shown in Fig. 6(a).

4) Mode 4 ($t_4 < t < t_5$, Fig. 3(d)): Mode 4 operates similarly to the positive cycle. Switch S_2 turns on at t_4 and S_1 remains off. Transformer $N_{56} - N_{78}$ stores energy in its magnetizing inductance. Its current $i_{L_{m_2}}$ and the corresponding flux densities B''_{o_1}, B''_c, and B''_{o_2} linearly increase. Simultaneously, transformer $N_{12} - N_{34}$ completes its discharge phase, with $i_{L_{m_1}}$ and i_{D_1} decreasing to zero at t_5.
5) Mode 5 ($t_5 < t < t_6$, Fig. 3(e)): Mode 5 starts after the discharge of transformer $N_{12} - N_{34}$ completes. Now, with Switch S_2 still on, transformer $N_{56} - N_{78}$ continues storing energy until $i_{L_{m_2}}$ reaches its peak value at t_6. The peak primary current in the negative cycle $i_{S_{2_{pk}}}$ is

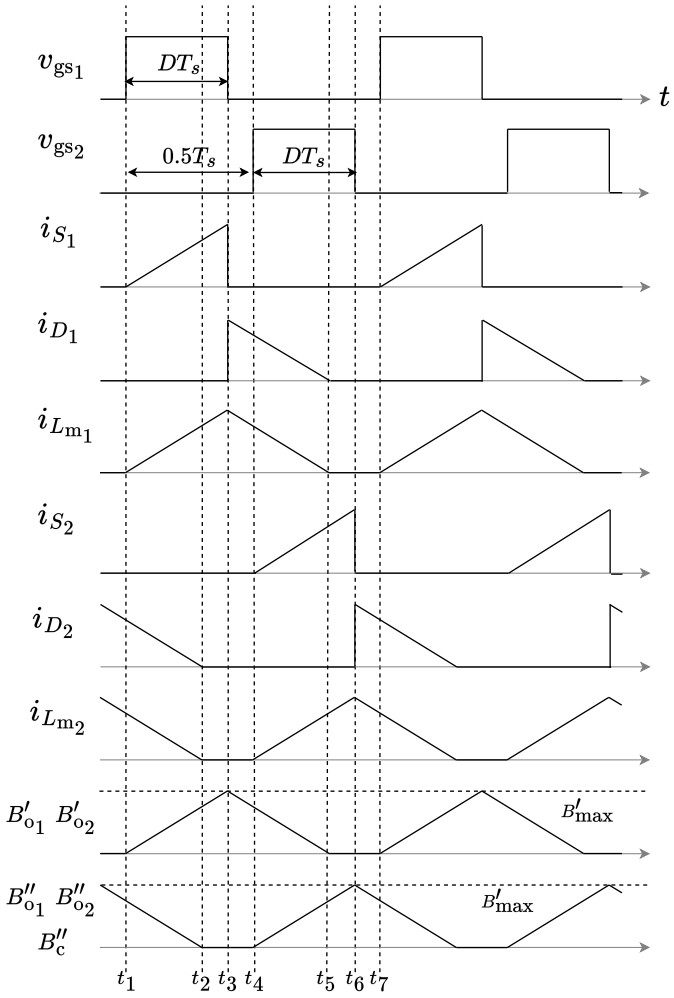


Fig. 4. Key waveforms in the operation of the two-phase interleaved flyback rectifier with the proposed transformer, with $D \leq 0.5\frac{nV_o}{V_{in_{max}}}$.

given by (4):

$$i_{S_{2_{pk}}} = \frac{V_{in} \cdot DT_s}{L_{m_2}} \tag{4}$$

where L_{m_2} is the magnetizing inductance of transformer $N_{56} - N_{78}$. Consequently, the flux densities B''_{o_1}, B''_c, and B''_{o_2} also reach their peak values. The peak value of $B''_{c_{pk}}$ is given by (5):

$$B''_{c_{pk}} = B''_{o_{pk}} = \frac{L_{m_2} i_{S_{2_{pk}}}}{N_{56} A_c} \tag{5}$$

with N_{56} as the number of turns in the primary side of the active transformer and A_c is the cross-sectional area of the center leg where the transformer $N_{56} - N_{78}$ is wound.

6) Mode 6 ($t_6 < t < t_7$, Fig. 3(f)): Switch S_2 turns off, initiating the discharge phase of transformer $N_{56} - N_{78}$. The currents $i_{L_{m_2}}, i_{D_2}$, and flux densities B''_{o_1}, B''_c, and B''_{o_2} begin to decrease linearly. Transformer $N_{12} - N_{34}$ remains inactive.

For Modes 4-6, since the magnetic resistance of both outer legs is equal, the magnetic flux is divided equally into both outer legs. Furthermore, since the number of turns of both outer legs is equal and wound in an opposite polarity, the electromotive force generated in $N_{12} - N_{34}$ is canceled. Therefore, the transformer formed by the center leg operates independently from the transformer in the outer leg, and the integrated transformer is equivalent to that of Fig. 5(c). The equivalent magnetic circuit for this case is shown in Fig. 6(b).

The proposed winding structure achieves zero magnetic coupling between the two transformers. Consequently, the distinct magnetic circuits share a single core without magnetic interference. Each transformer maintains independent operation despite the physical integration. This structure eliminates the need for two discrete magnetic components. As a result, the total volume of the magnetic components is significantly reduced.

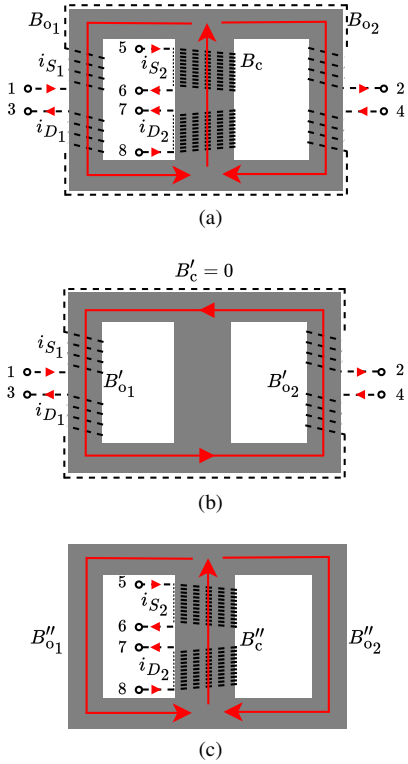


Fig. 5. The proposed integrated transformer during the two phases of the converter, with the flux density components in each leg illustrated in (a). (b) is realized during the positive AC cycle, while (c) is realized during the negative cycle.

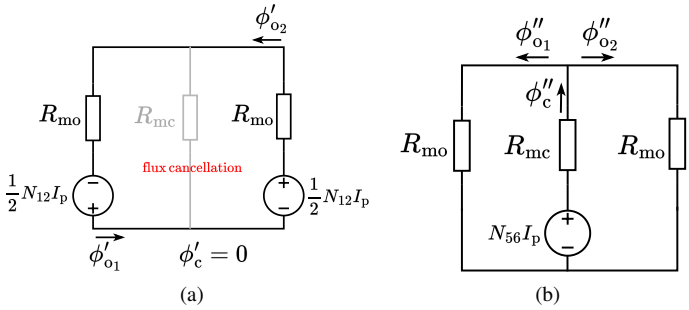


Fig. 6. Equivalent magnetic circuits for the proposed transformer. (a) shows the equivalent magnetic circuit for $N_{12}-N_{34}$, while (b) shows the equivalent magnetic circuit for $N_{56} - N_{78}$.

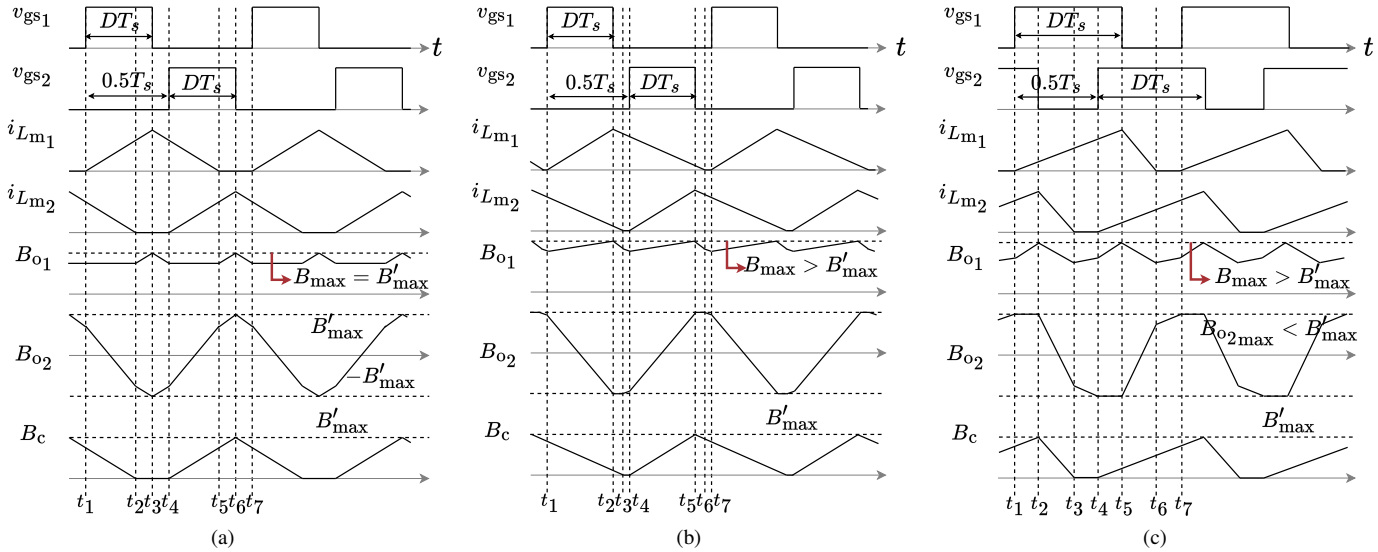


Fig. 7. Theoretical, superimposed magnetic flux density waveforms in the integrated transformer during circuit operations: (a) Case 1, (b) Case 2, and (c) Case 3.

B. Saturation Magnetic Flux Density Analysis

The circuit operation is classified into three patterns determined by D and the boundary condition $D \leq 0.5\frac{nV_o}{V_{in_{max}}}$. This classification, illustrated in Fig. 7, arises because the integrated single-core structure permits simultaneous magnetizing current flow in multiple active windings, resulting in distinct flux superposition compared to conventional discrete transformers. The manner of superposition due to the magnetizing current can be controlled by t_{dis}. Consequently, the saturation magnetic flux density B_{max} is a critical parameter for the transformer geometry. It can be expressed in terms of the saturation magnetic flux density in each phase, denoted as B'_{max} and B''_{max}, with $B'_{max} = B''_{max}$ since $L_{m_1} = L_{m_2}$. The flux density in each leg is examined further using the principle of superposition, with the expressions of the superimposed flux densities in the left, right and center legs shown in (6)-(8), represented as B_{o_1}, B_{o_2} and B_c respectively:

$$B_{o_1} = B'_{o_1} + B''_{o_1} \tag{6}$$

$$B_{o_2} = B'_{o_2} - B''_{o_2} \tag{7}$$

$$B_c = B''_c \tag{8}$$

Case 1: $D \leq 0.5\frac{nV_o}{V_{in_{max}}}$*:* The waveforms in Fig. 4 represent the circuit operation for Case 1. As shown, flux cancellation does not occur in the outer legs; however, the magnitude of the peak flux density remains uniform across all legs. If the magnetizing current of one phase reaches zero before the magnetizing current of the other phase reaches its peak, then $B_{max} = B'_{max}$. This condition is satisfied when D is expressed by (9).

$$D \leq 0.5\frac{nV_o}{V_{in_{max}}} \tag{9}$$

Case 2: $0.5\frac{nV_o}{V_{in_{max}}} < D \leq 0.5\frac{nV_o}{V_{in_{max}}}$*:* In this case, the magnetizing current of one phase does not reach zero before the subsequent phase reaches its peak. The saturation flux density across all phases can be derived as follows. At t_2, $B'_{o_1} = B'_{max}$. Moreover, $B''_{o_1}(t_2) = B'_{o_1}(t_5)$ and is expressed in (10):

$$B''_{o_1}(t_2) = B'_{o_1}(t_5) = B'_{max} - \frac{B'_{max}}{t_6 - t_2}(t_5 - t_2) \tag{10}$$

with $t_5 = 0.5T_s + t_2$ and $t_2 = DT_s$. The peak $i_{Lm_{1pk}}$ can be rewritten as in (11):

$$i_{Lm_{1pk}} = \frac{V_{in_{max}}}{L_m} DT_s = \frac{nV_o}{L_m}(t_6 - t_2) \tag{11}$$

In this case, (9) is not satisfied, and this results in B_{max} in the left outer leg exceeding B'_{max} in the center leg, as expressed in (12):

$$B_{max} = \left(2 - \frac{0.5nV_o}{V_{in_{max}} D}\right) B'_{max} \tag{12}$$

Case 3: $0.5 < D$*:* In this case, B_{max} in the left outer leg consistently exceeds B'_{max}. By analysis similar to Case 2, B_{max} is derived as (13):

$$B_{max} = \left(1 + \frac{D - 0.5}{D}\right) B'_{max} \tag{13}$$

Because the flyback is interleaved, it is expected that the rectifier will primarily operate at $D \leq 0.5$, which operates in the conditions set by Case 1 or 2.

C. Design Method of the Proposed Integrated Transformer

A turns ratio n is selected in consideration of the maximum voltage ratings of the switches. The maximum operating duty cycle D is selected. The magnetizing inductance, $L_{m_1} = L_{m_2} = L_m$ is calculated from the circuit specifications. The saturation magnetic flux density per phase B'_{max} is calculated from the B_{max} given by the chosen core material, considering the maximum D. A core is chosen using the area-product (AP) method.

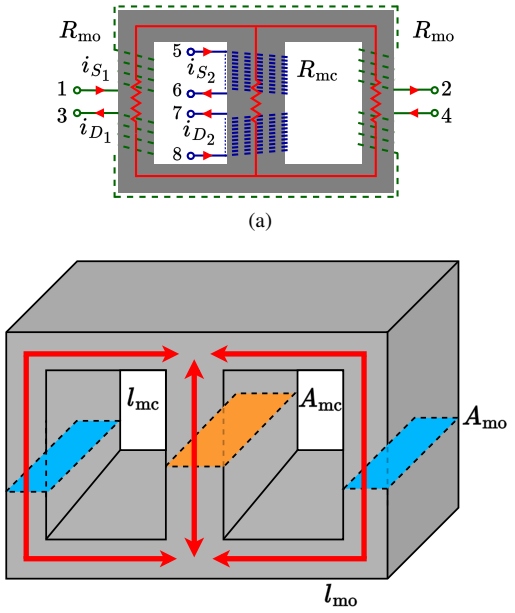


Fig. 8. EE core configuration: (a) magnetic resistances of the transformer (R_{mo} for the outer legs, R_{mc} for the center leg), and (b) effective cross-sectional area (A_{mo} for the outer legs, A_{mc} for the center leg) and effective magnetic path length l_{mo} for the outer legs, l_{mc} for the center leg.

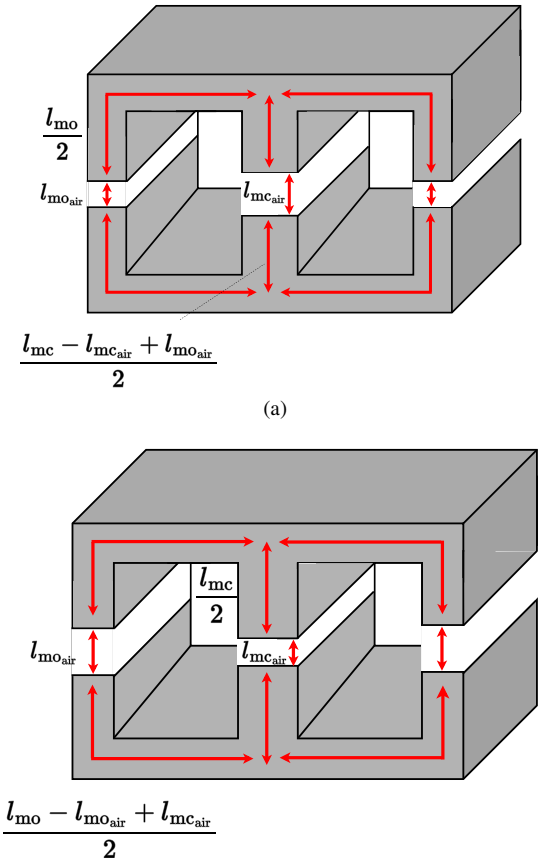


Fig. 9. EE core configuration with air gaps: (a) setup when $l_{\mathrm{mo_{air}}} > l_{\mathrm{mc_{air}}}$ and (b) setup when $l_{\mathrm{mo_{air}}} < l_{\mathrm{mc_{air}}}$.

The number of turns is calculated using $B'_{\max}$ and the peak primary side current $I_{\mathrm{pri_{pk}}}$ as shown in (14) and (15):

$$\frac{N_{12}}{n} = N_{34} \geq \frac{L_{\mathrm{m}} I_{\mathrm{pri_{pk}}}}{n A_{\mathrm{mo}} B'_{\max}} \tag{14}$$

$$\frac{N_{56}}{n} = N_{78} \geq \frac{L_{\mathrm{m}} I_{\mathrm{pri_{pk}}}}{n A_{\mathrm{mc}} B'_{\max}} \tag{15}$$

The magnetic resistances of the outer legs R_{mo} and the center leg R_{mc} are obtained from a magnetic circuit analysis of the integrated transformer in its separat phases. The transformer $N_{12} - N_{34}$ is only made of the outer legs as shown in Fig. 8(b), while $N_{56} - N_{78}$ is made of all the legs. The magnetizing inductance L_{m} in terms of the number of turns and magnetic resistances for $N_{12} - N_{34}$ and $N_{56} - N_{78}$ are shown in (16) and (17) respectively.

$$L_{\mathrm{m}} = \frac{N_{12}^2}{2R_{\mathrm{mo}}} \tag{16}$$

$$L_{\mathrm{m}} = \frac{N_{56}^2}{R_{\mathrm{mc}} + 0.5R_{\mathrm{mo}}} \tag{17}$$

The magnetic resistances R_{mo} and R_{mc} are obtained from (18) and (19):

$$R_{\mathrm{mo}} = \frac{N_{12}^2}{2L_{\mathrm{m}}} \tag{18}$$

$$R_{\mathrm{mc}} = \frac{(2N_{56} - N_{12})(2N_{56} + N_{12})}{4L_{\mathrm{m}}} \tag{19}$$

Finally, an air gap was also inserted in each of the legs to provide the desired resistance. Non-equal length air gaps were placed in the transformer. There are two cases, one where the air gaps in the outer legs $l_{\mathrm{mo_{air}}}$ are longer than the air gap in the center leg $l_{\mathrm{mc_{air}}}$, and one where $l_{\mathrm{mc_{air}}}$ is longer than $l_{\mathrm{mo_{air}}}$.

Case 1: $l_{\mathrm{mo_{air}}} > l_{\mathrm{mc_{air}}}$: Given the setup in Fig. 9(a), R_{mo} and R_{mc} are shown in (20) and (21) respectively with $l_{\mathrm{mo_{air}}}$ and $l_{\mathrm{mc_{air}}}$ in (22) and (23) respectively.

$$R_{\mathrm{mo}} = \frac{l_{\mathrm{mo}} - l_{\mathrm{mo_{air}}} + l_{\mathrm{mc_{air}}}}{\mu_r \mu_0 A_{\mathrm{mo}}} + \frac{l_{\mathrm{mo_{air}}}}{\mu_0 A_{\mathrm{mo}}} \tag{20}$$

$$R_{\mathrm{mc}} = \frac{l_{\mathrm{mc}}}{\mu_r \mu_0 A_{\mathrm{mc}}} + \frac{l_{\mathrm{mc_{air}}}}{\mu_0 A_{\mathrm{mc}}} \tag{21}$$

$$l_{\mathrm{mo_{air}}} = \frac{\mu_0(\mu_r A_{\mathrm{mo}} R_{\mathrm{mo}} - A_{\mathrm{mc}} R_{\mathrm{mc}}) + \frac{l_{\mathrm{mc}}}{\mu_r} - l_{\mathrm{mo}}}{\mu_r - 1} \tag{22}$$

$$l_{\mathrm{mc_{air}}} = \mu_0 A_{\mathrm{mc}} R_{\mathrm{mc}} - \frac{l_{\mathrm{mc}}}{\mu_r} \tag{23}$$

Case 2: $l_{\mathrm{mo_{air}}} < l_{\mathrm{mc_{air}}}$: Given the setup in Fig. 9(b), R_{mo} and R_{mc} are shown in (24) and (25) respectievly $l_{\mathrm{mo_{air}}}$ and $l_{\mathrm{mc_{air}}}$ in (26) and (27) respectively.

$$R_{\mathrm{mo}} = \frac{l_{\mathrm{mo}}}{\mu_r \mu_0 A_{\mathrm{mo}}} + \frac{l_{\mathrm{mo_{air}}}}{\mu_0 A_{\mathrm{mo}}} \tag{24}$$

$$R_{\mathrm{mc}} = \frac{l_{\mathrm{mc}} - l_{\mathrm{mc_{air}}} + l_{\mathrm{mo_{air}}}}{\mu_r \mu_0 A_{\mathrm{mc}}} + \frac{l_{\mathrm{mc_{air}}}}{\mu_0 A_{\mathrm{mc}}} \tag{25}$$

$$l_{\mathrm{mo_{air}}} = \mu_0 A_{\mathrm{mo}} R_{\mathrm{mo}} - \frac{l_{\mathrm{mo}}}{\mu_r} \tag{26}$$

$$l_{\mathrm{mc_{air}}} = \frac{\mu_0(\mu_r A_{\mathrm{mc}} R_{\mathrm{mc}} - A_{\mathrm{mo}} R_{\mathrm{mo}}) + \frac{l_{\mathrm{mo}}}{\mu_r} - l_{\mathrm{mc}}}{\mu_r - 1} \tag{27}$$

III. Simulation Verification of Proposed Design

A. Simulation Verification of the Proposed Design Method

To ensure the valid operation of the proposed integrated transformer, a simulation of the two-phase interleaved flyback rectifier was first performed in PLECS Standalone 4.9.4, with the diagram shown in Fig. 10(a). Clamped RCD Snubbers ($C_{\mathrm{sn}_1}, D_{\mathrm{sn}_1}, R_{\mathrm{sn}_1}$ and $C_{\mathrm{sn}_2}, D_{\mathrm{sn}_2}, R_{\mathrm{sn}_2}$) were also added for protection against surge voltage [15], [16] caused by the leakage inductance and the switch capacitance at the turn-off of S_1 and S_2. An integrated transformer, shown in Fig. 10(b), was also designed using the proposed method. The circuit and transformer were designed with the condition given by (9), and will thus be verified in this study. The specifications of the two-phase interleaved flyback rectifier and transformer for the simulation verification are shown in Tables II and III respectively. The duty cycle is calculated using (28) [8].

$$D = \frac{2}{V_{\mathrm{in_{pk}}}} \sqrt{\frac{0.5 L_{\mathrm{m}} P_{\mathrm{o}}}{T_{\mathrm{s}}}} \tag{28}$$

B. Simulation Results

Fig. 11 shows waveforms from the simulation testing of the two-phase interleaved flyback rectifier with the proposed transformer at 240 W. Key waveforms of the circuit were observed from the simulation, which include v_{gs_1} and v_{gs_2}, $i_{L_{\mathrm{m}_1}}$ and $i_{L_{\mathrm{m}_2}}$, and the fluxes B_{o_1}, B_{o_2}, and B_{c}. Similar waveforms to that of Fig. 4 have been observed. Flux cancellation is observed in the center leg during the time

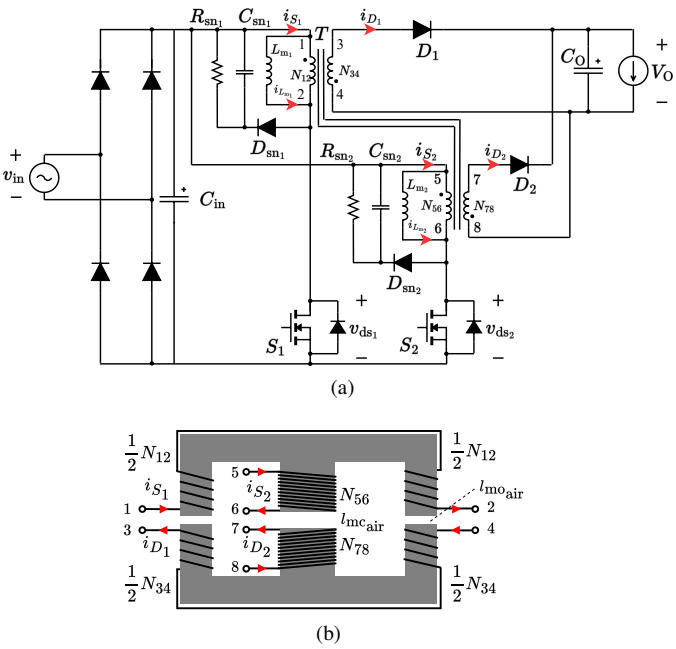


Fig. 10. Prototype two-phase interleaved flyback rectifier (a) with the proposed integrated transformer (b).

that $N_{56} - N_{78}$ is inactive, as denoted by the interval when $i_{L_{\mathrm{m}_2}} = 0$. The magnetizing currents $i_{L_{\mathrm{m}_1}}$ and $i_{L_{\mathrm{m}_2}}$ reach zero before one other phase reaches its peak. Moreover, the increase and decrease of fluxes were all linear, therefore confirming that there is indeed no magnetic coupling interference. This confirms the validity of the theoretical operation proposed in Section II.

TABLE II
Simulation Specifications of the Prototype Two-Phase Interleaved Flyback Rectifier

Item	Symbol	Value
AC input voltage	v_{in}	100 V V_{rms}
Maximum rated power	P_{O}	240 W
Output voltage	V_{O}	48 V
Switching frequency	f_{s}	100 kHz
Input capacitor	C_{in}	1 µF
Output capacitor	C_{O}	2 x 1mF (parallel)
Snubber capacitor	$C_{\mathrm{sn}_1}, C_{\mathrm{sn}_2}$	2.2 nF
Snubber resistor	$R_{\mathrm{sn}_1}, R_{\mathrm{sn}_2}$	30 kΩ

TABLE III
Simulation Specifications of the Prototype Integrated Transformer

Item	Symbol	Value
Magnetic core	-	OP44721EC
Relative permeability	μ_r	2500
Magnetic path length (center leg)	l_{mc}	31.7 mm
Magnetic path length (outer leg)	l_{mo}	71.4 mm
Cross-sectional area (center leg)	A_{mc}	243.4 mm^2
Cross-sectional area (outer leg)	A_{mo}	117.6 mm^2
Magnetizing inductance	$L_{\mathrm{m}_1}, L_{\mathrm{m}_2}$	82.8 µH
Turn ratio	n	3
Number of turns	$N_{12} : N_{34}$	24:8
	$N_{56} : N_{78}$	18:6
Center leg air gap	$l_{\mathrm{mc_{air}}}$	0.650 mm
Outer leg air gap	$l_{\mathrm{mo_{air}}}$	0.500 mm
Coupling coefficient of N_{12}, N_{34}	$k_{N_{12}-N_{34}}$	0.990
Coupling coefficient of N_{56}, N_{78}	$k_{N_{56}-N_{78}}$	0.990
Coupling coefficient of N_{12}, N_{56}	$k_{N_{12}-N_{56}}$	0.000

IV. Size Reduction Effect of the Proposed Transformer

This section examines the size reduction effect by the integrated transformer. For this study, the conventional transformer

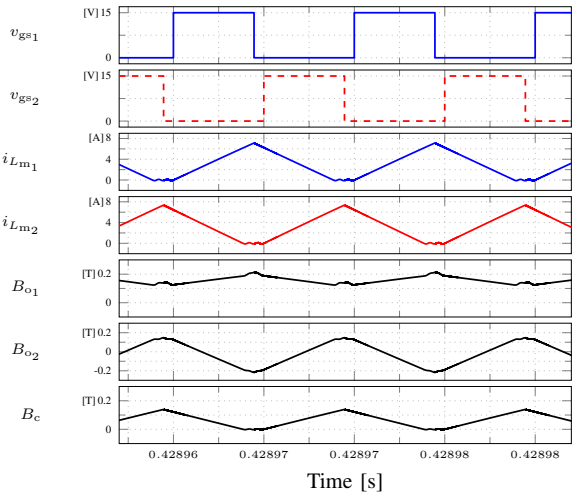


Fig. 11. Key simulation waveforms of the proposed two-phase interleaved flyback rectifier at 240 W.

is assumed to be two separate transformers that facilitate isolation between the primary and secondary side. A representation is shown in Fig. 12(a), and its windings are represented in Fig. 13(a). The proposed integrated transformer is represented in Fig. 12(b) and its windings are represented in Fig. 13(b). The volume is compared by using the AP method. The AP method utilizes the product between the core cross-sectional area, A_{c}, and the window-area, A_{w}, and is a fundamental core size for the application [21]. It can be expressed both in terms of the transformer geometric parameters and the circuit parameters, as will be demonstrated in the following analysis.

The EE-core was redesigned in such a way such that A_{w} and A_{c} are squares to allow for simpler analysis, as shown in Fig. 14 [22]. Here, A_{c} is twice the area of the outer leg.

For the conventional transformer, A_{c} is expressed in (29). Here, N_{p} is the number of turns wound on the center leg of the conventional transformer.

$$A_{\mathrm{c_{conv}}} = \frac{L_{\mathrm{m}} I_{\mathrm{pri_{pk}}}}{B_{\max} N_{\mathrm{p}}} \tag{29}$$

For the integrated transformer A_{c} is expressed in (30).

$$A_{\mathrm{c_{integ}}} = \frac{L_{\mathrm{m}} I_{\mathrm{pri_{pk}}}}{B_{\max} N_{56}} \tag{30}$$

The window area A_{w} can be expressed in terms of winding area and the occupancy ratio K_{u}, as seen in (31) for the conventional transformer:

$$A_{\mathrm{w_{conv}}} K_{\mathrm{u}} = N_{\mathrm{p}} \frac{I_{\mathrm{pri_{rms}}}}{J_{\mathrm{m}}} + N_{\mathrm{s}} \frac{I_{\mathrm{sec_{rms}}}}{J_{\mathrm{m}}} \tag{31}$$

which simplifies to:

$$A_{\mathrm{w_{conv}}} = \frac{N_{\mathrm{p}}}{J_{\mathrm{m}} K_{\mathrm{u}}} \left(I_{\mathrm{pri_{rms}}} + \frac{I_{\mathrm{sec_{rms}}}}{n} \right) \tag{32}$$

For the integrated transformer, taking into account that the core window has half the total turns of the transformer N_{12} and N_{34}, the A_{w} can be expressed as seen in (33).

$$A_{\mathrm{w_{integ}}} K_{\mathrm{u}} = \left(N_{56} \frac{I_{\mathrm{pri_{rms}}}}{J_{\mathrm{m}}} + N_{78} \frac{I_{\mathrm{sec_{rms}}}}{J_{\mathrm{m}}} \right) + \frac{1}{2} \left(N_{12} \frac{I_{\mathrm{pri_{rms}}}}{J_{\mathrm{m}}} + N_{34} \frac{I_{\mathrm{sec_{rms}}}}{J_{\mathrm{m}}} \right) \tag{33}$$

The total volume of the EE core in Fig. 14 expressed in (34), where x is a constant that denotes the ratio $A_{\mathrm{c}}/A_{\mathrm{w}}$. The volume corresponding to the conventional and integrated transformer is obtained using their respective A_{c} and A_{w}.

$$\mathrm{Vol} = 2AP^{\frac{3}{4}}(2x^{\frac{1}{4}} + x^{\frac{3}{4}} + x^{-\frac{1}{4}}) \tag{34}$$

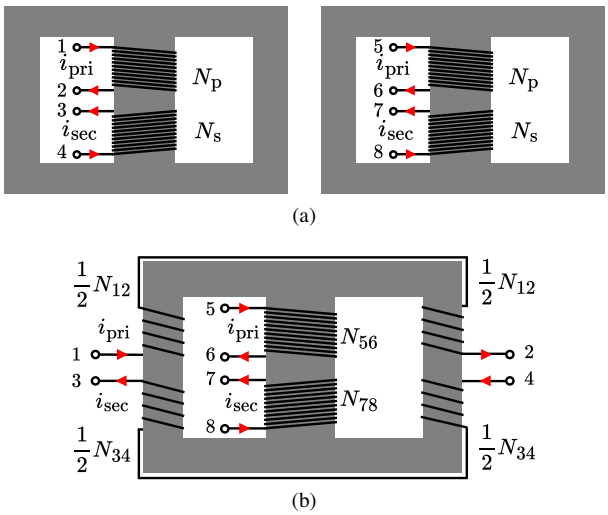


Fig. 12. Two separate transformers (a) and proposed integrated transformer (b) used in the volume analysis.

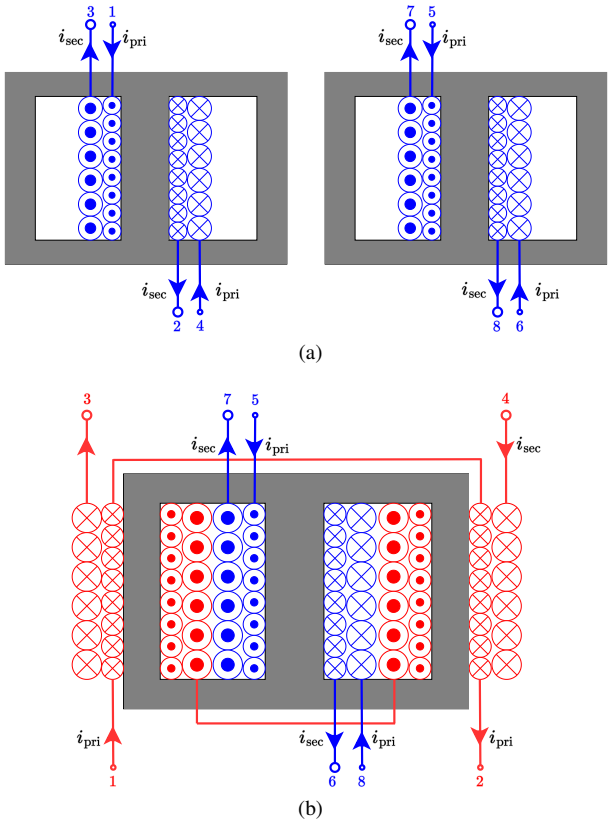


Fig. 13. Two separate transformers (a) and the proposed integrated transformer (b) used in the volume analysis, with the windings represented as well.

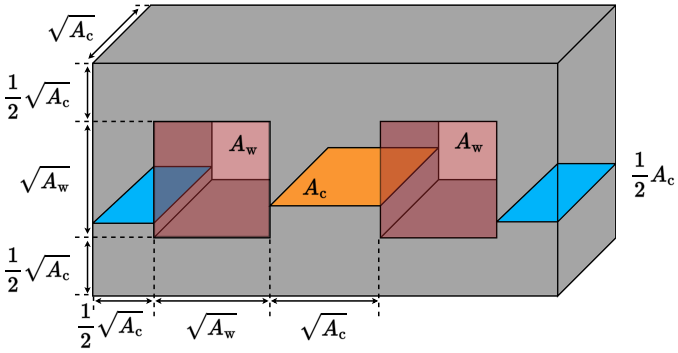


Fig. 14. Simplified representation of one EE-core used in the volume analysis.

The rms expressions of the primary and secondary side current are given in (35) and (36) [8].

$$I_{\mathrm{pri}_{\mathrm{rms}}} = \frac{V_{\mathrm{in}_{\mathrm{max}}}}{L_{\mathrm{m}} f_{\mathrm{s}}} \sqrt{\frac{D^3}{6}} \tag{35}$$

$$I_{\mathrm{sec}_{\mathrm{rms}}} = \frac{V_{\mathrm{in}_{\mathrm{max}}}}{L_{\mathrm{m}} f_{\mathrm{s}}} \sqrt{\frac{4nD^3}{9\pi} \cdot \frac{V_{\mathrm{in}_{\mathrm{max}}}}{V_{\mathrm{o}}}} \tag{36}$$

Using these equations with the actual circuit specfications, a 56% volume reduction has been achieved by the proposed design, with the comparison shown in Table IV. This assumes that the integrated core geometry is custom-tailored to the specific area product requirements derived from the circuit specifications, rather than constrained by standard commercial core shapes. While the experimental prototype described in Section V utilizes a commercial core (Magnetics 0P44721EC) for proof-of-concept, this theoretical analysis confirms that a purpose-built integrated core would offer significant improvements in power density compared to the conventional discrete implementation.

V. Experimental Verification of Proposed Design

A. Prototype Evaluation Circuit and Prototype Transformer

A prototype two-phase interleaved flyback rectifier, shown in Fig. 10(a) operating in DCM was built to verify the operation of the integrated transformer. The prototype integrated transformer, shown in Fig. 10(b), was also designed using the proposed method. The specifications of the two-phase interleaved flyback rectifier and transformer are shown in Tables V and VI respectively. The transformer parameters were characterized using a Hioki IM3536 LCR meter at 100 kHz. The coupling coefficients of the transformers $k_{N_{12}-N_{34}}$, $k_{N_{56}-N_{78}}$, and $k_{N_{12}-N_{56}}$ were derived from the mutual inductance obtained after connecting the respective transformer windings in series-aiding and series-opposing configurations. The coupling coefficient between the two flyback transformers $k_{N_{12}-N_{56}}$ is 0.0005, showing that the coupling was almost eliminated.

TABLE IV
Volume Analysis of the Conventional and Proposed Transformer

Item	Symbol	Conventional	Proposed
Maximum input voltage	$V_{\mathrm{in}_{\mathrm{max}}}$	240 V V_{rms}	
Output voltage	$V_{\mathrm{in}_{\mathrm{max}}}$	48 V	
Maximum duty cycle	D	0.47	
Switching frequency	f_{s}	100 kHz	
Magnetizing inductance	L_{m}	82.8 µH	
Turn ratio	n	3	
Number of turns (Conv.)	$N_{\mathrm{p}} : N_{\mathrm{s}}$	24:8	–
Number of turns (Prop.)	$N_{12} : N_{34}$	–	24:8
	$N_{56} : N_{78}$	–	18:6
Primary side peak current	$I_{\mathrm{pri,pk}}$	8.02 A	
Primary side rms current	$I_{\mathrm{pri,rms}}$	5.39 A	
Secondary side rms current	$I_{\mathrm{sec,rms}}$	12.53 A	
Saturation flux density	B_{max}	0.3 T	
Maximum current density	J_{m}	5 A/mm^2	
Occupancy ratio	K_{u}	0.3	
Area Product	AP	14128.75 mm^4	23547.91 mm^4
Core Volume	Vol	21106.38 mm^3	9398.46 mm^3

TABLE V
Specifications of the Prototype Two-Phase Interleaved Flyback Rectifier

Item	Symbol	Value / Component
AC input voltage	v_{in}	100-240 V V_{rms}
Maximum rated power	P_{O}	240 W
Output voltage	V_{O}	48 V
Switching frequency	f_{s}	100 kHz
Bridge diode	BD	CD-DF408S
Input capacitor	C_{in}	1 µF
Switch	S_1, S_2	STF15N80k5
Secondary diode	D_1, D_2	STPSC30H12CWL
Output capacitor	C_{O}	2 x 1mF, in parallel
Snubber capacitor	$C_{\mathrm{sn}_1}, C_{\mathrm{sn}_2}$	2.2 nF
Snubber resistor	$R_{\mathrm{sn}_1}, R_{\mathrm{sn}_2}$	30 kΩ
Snubber diode	$D_{\mathrm{sn}_1}, D_{\mathrm{sn}_2}$	VS-E7MH0112-M3/1

TABLE VI
Specifications of the Prototype Integrated Transformer

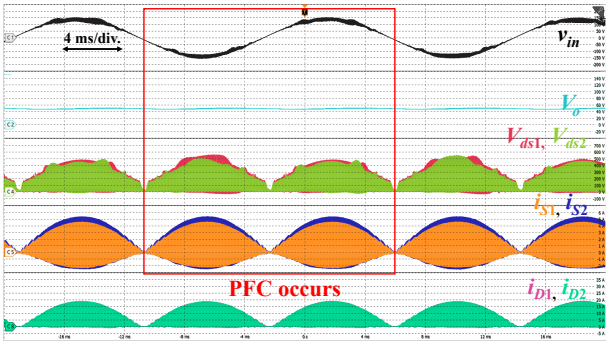
Item	Symbol	Value
Magnetizing inductance (Target)	L_{m}	82.8 µH
Magnetizing inductance of N_{12}	L_{m1}	86.1 µH
Magnetizing inductance of N_{56}	L_{m2}	75.1 µH
Turn ratio	n	3
Magnetic core	-	OP44721EC
Number of turns	$N_{12} : N_{34}$	24:8
	$N_{56} : N_{78}$	15:5
Center leg air gap	$l_{\mathrm{mc}_{\mathrm{air}}}$	0.650 mm
Outer leg air gap	$l_{\mathrm{mo}_{\mathrm{air}}}$	0.500 mm
Coupling coefficient of N_{12}, N_{34}	$k_{N_{12}-N_{34}}$	0.970
Coupling coefficient of N_{56}, N_{78}	$k_{N_{56}-N_{78}}$	0.979
Coupling coefficient of N_{12}, N_{56}	$k_{N_{12}-N_{56}}$	0.0005

B. Experimental Results

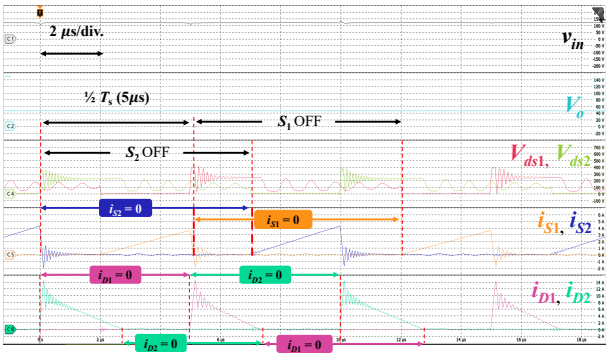
Fig. 15(a) shows the 100 W experimental waveforms at 4 ms/div from the two-phase interleaved flyback rectifier with the proposed transformer, demonstrating power factor correction. Fig. 15(b) presents 2 µs/div waveforms. No current is induced in either switch except during the resonance between the leakage inductance and the parasitic capacitance of the switch. During the inactive period of the transformer, secondary diode side current is zero. Moreover, after i_{S_1} and i_{S_2} reached zero when S_1 and S_2 switched off, the currents remained at zero. Therefore, the two flyback converters operate independently without magnetic coupling interference. The same phenomenon was observed at 240 W two-phase drive, as shown in Fig. 15(c). Single-phase drive at lower power at 32.5 W is also feasible as shown in Fig. 15(d).

VI. Conclusion

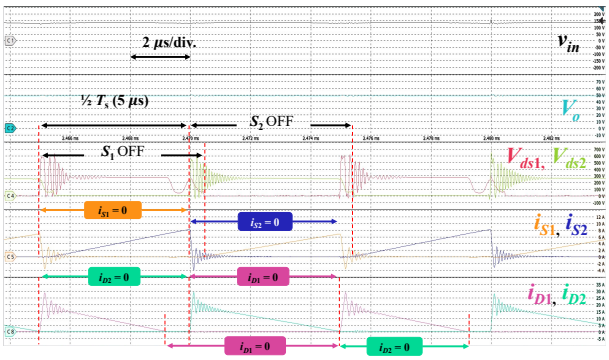
In this paper, an integrated transformer was successfully designed for a two-phase interleaved flyback rectifier for USB PD 3.1 applications. The proposed winding structure for the integrated transformer prevents coupling between the two flyback transformers. A coupling coefficient of 0.0005 was achieved with the prototype. Independent flyback circuit operation without magnetic interference due to coupling was verified through simulation and experiment up to 240 W. Because of the proposed design, two transformers are integrated into one component, enabling the downsizing of magnetic components by about 19% compared to the conventional transformer.



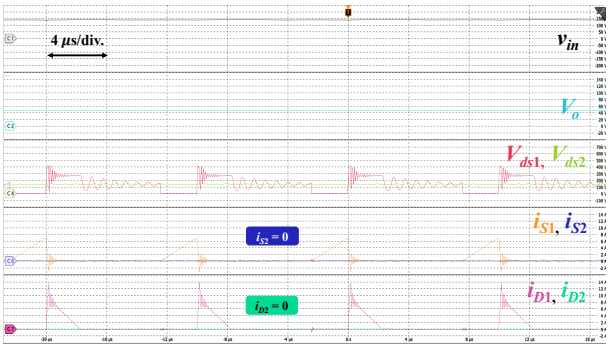
(a)



(b)



(c)



(d)

Fig. 15. Experimental waveforms of a two-phase interleaved flyback rectifier with the proposed integrated transformer at (a) & (b): 100 W, two-phase AC drive, (c): 240 W, two-phase AC drive, (d): 32.5 W, single-phase AC drive. For (a) & (b): $v_{\rm in}$: 50 V/div., $V_{\rm o}$: 10 V/div., $v_{\rm ds_1}, v_{\rm ds_2}$: 100 V/div., $i_{\rm S_1}, i_{\rm S_2}$: 1 A/div., $i_{\rm D_1}, i_{\rm D_2}$: 5 A/div.; For (c): $v_{\rm in}$: 50 V/div., $V_{\rm o}$: 10 V/div., $v_{\rm ds1}, v_{\rm ds2}$: 100 V/div., $i_{\rm S_1}, i_{\rm S_2}$: 2 A/div., $i_{\rm D_1}, i_{\rm D_2}$: 5 A/div.; For (d): $v_{\rm in}$: 50 V/div., $V_{\rm o}$: 20 V/div., $v_{\rm ds_1}, v_{\rm ds_2}$: 100 V/div., $i_{\rm S_1}, i_{\rm S_2}$: 2 A/div., $i_{\rm D_1}, i_{\rm D_2}$: 2 A/div.

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